

Temporal multimodal fusion with AI-enabled textual and macroeconomic features for multi-horizon oil price prediction

Yen-Wen Chen , Welbey Prasadirta, Chia-Yen Lee *

Department of Information Management, National Taiwan University, Taipei 106, Taiwan

ARTICLE INFO

Keywords:

Crude oil price forecasting
Multimodal fusion
Large language models (LLMs)
Time series decomposition
Macroeconomic indicators

ABSTRACT

Accurate multi-horizon crude oil price forecasting is difficult because market dynamics are nonlinear, nonstationary, and influenced by both numerical fundamentals and event-driven text. This study proposes a decomposition-based multimodal framework that combines large-scale macroeconomic indicators with LLM-enhanced news features for West Texas Intermediate (WTI) forecasting. Full-length financial news is transformed into structured summaries, keywords, and headlines using GPT-4.1, then encoded with FinBERT embeddings. In parallel, numerical predictors are screened via Granger-causality tests, and WTI prices are decomposed into trend, seasonal, and residual components using seasonal-trend decomposition (STL). We evaluate early, intermediate, and late fusion under an expanding-window protocol across 1- to 8-week horizons. Results indicate that early fusion achieves the lowest average errors across horizons (e.g., Week-8 MAPE 5.78%, Week-8 MSE 23.59) while maintaining competitive fold-level stability. At the 4-week horizon, the model attains MAPE 3.45%, MSE 6.81, and trend accuracy 81.51%. These findings support the value of integrating LLM-derived textual context and macroeconomic signals in a component-aware architecture for practical medium-term crude oil forecasting.

1. Introduction

Crude oil is a cornerstone of the global economy, and price swings shape inflation, industrial output, and energy policy (Patidar et al., 2024). Recent shocks such as the Coronavirus Disease 2019 (COVID-19) pandemic and geopolitical conflicts highlight the market's extreme volatility and the need for reliable forecasts (Bilal et al., 2024). These dynamics are nonlinear and nonstationary, with jumps driven by supply-demand imbalances, geopolitics, and speculation (Selmi et al., 2023).

Recent forecasting work increasingly adopts data-driven models and decomposition schemes. Deep learning combined with signal decomposition, such as variational mode decomposition (VMD) and empirical mode decomposition (EMD) frameworks, can improve accuracy by isolating latent components (Fang et al., 2023a; Huang & Deng, 2021). Seasonal-Trend decomposition using Loess (STL) approaches further separate trend, seasonal, and residual behavior and allow different predictors to act at different frequencies (Ding et al., 2024). However, many models still rely primarily on numerical time series and do not fully exploit the information contained in unstructured text.

Textual information from news and online sources provides early signals for price movements, and prior studies show that text features complement market indicators in crude oil forecasting (Fang et al., 2023b; Li et al., 2019; Wu et al., 2021). Yet most approaches reduce text to

headline-level sentiment or polarity, which can miss contextual cues and event structure. In addition, decomposition components are often treated as black boxes, and multimodal fusion is frequently performed late in the modeling pipeline, limiting cross-modal interactions. Finally, many studies emphasize short horizons, while procurement and planning decisions require multi-week visibility (Chen et al., 2015).

This emphasis on multi-week forecasting also follows procurement practice. In many energy-intensive firms, purchase contracts are often committed several weeks ahead, and inventory adjustments are planned on weekly or monthly cycles under shipping lead times and logistics constraints. The 1–8 week horizon is therefore chosen to align model outputs with actionable planning windows, where anticipated price paths can inform advance purchasing, inventory rebalancing, contract timing, and short-term hedging decisions. The three STL components align naturally with these decision layers. The trend component, driven by macroeconomic fundamentals, supports the recurring choice between committing to a fixed-price supply agreement and maintaining floating exposure, a decision whose value depends directly on whether structural prices are rising or falling over the relevant lead time (Lee et al., 2022). Inventory planning operates at the seasonal layer: the 28-day demand cycle that STL isolates, tied to monthly inventory releases, contract renewals, and shipping schedules, provides the timing signal that buffer stock adjustments require. News-driven shocks, including sup-

* Corresponding author.

E-mail addresses: d12725002@ntu.edu.tw (Y.-W. Chen), r12725060@ntu.edu.tw (W. Prasadirta), chiayenlee@ntu.edu.tw (C.-Y. Lee).

ply disruptions, geopolitical events, and unexpected inventory figures, concentrate their predictive content in the residual component, where LLM-enhanced text embeddings are most informative and support rapid hedging responses. The 81.51% directional accuracy at the four-week horizon translates into a measurable operational edge on the binary fixed-versus-floating contract decision that recurs each procurement cycle (Hsu et al., 2025).

To address these gaps, we propose a decomposition-based multimodal fusion framework for multi-horizon West Texas Intermediate (WTI) forecasting that couples macroeconomic indicators with context-rich news representations. Our contributions are:

- **LLM-enhanced feature engineering:** We use Generative Pre-trained Transformer (GPT) 4.1 (GPT-4.1)¹, a generative large language model (LLM), to transform full articles into structured features (alternative headline, keywords, and summary) that preserve context. We encode the LLM-enhanced text using FinBERT to capture nuanced financial semantics beyond polarity scores.
- **Decomposition-based multi-modal architecture:** We apply STL to decompose WTI into trend, seasonal, and residual components and assign tailored inputs to each component via dedicated long short-term memory (LSTM) networks for interpretability and accuracy.
- **Fusion strategy evaluation:** We systematically compare early, intermediate, and late fusion. The systematic evaluation identifies the most effective strategy for integrating numerical and textual data for crude oil forecasting, which is largely absent in prior literature. The results show that early fusion delivers the most stable and accurate long-horizon forecasts in this setting.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on crude oil price forecasting. Section 3 details the proposed multimodal fusion framework. Section 4 presents the experimental setup and discusses the results. Finally, Section 5 concludes the paper and outlines directions for future research.

2. Related works

Crude oil price forecasting has progressed from purely numerical models to approaches that combine decomposition, text signals, and multimodal learning. We organize prior work into three threads: (1) time-series and decomposition models, (2) textual signals and sentiment, and (3) multimodal fusion strategies.

2.1. Time-series and decomposition models

Early work focused on statistical and pattern-based forecasts, including multi-step pattern matching approaches for crude oil prices (Fan et al., 2008). To address nonstationarity, many studies adopt decomposition to isolate smoother components before modeling. VMD-based frameworks such as VMD-LSTM improve forecasting by decomposing the series into stationary subseries (Huang & Deng, 2021). Multifactor search combined with multivariate empirical mode decomposition (MEMD) also shows that multi-scale search data improves level and directional accuracy (Tang et al., 2020). EMD variants are also common; for example, Fang et al. combine an improved slope-based method (ISBM) with a feedforward neural network (FNN) in an EMD-based hybrid to forecast crude oil prices (Fang et al., 2023a). More recently, STL-based approaches provide interpretable trend, seasonal, and residual components and allow predictors to act at different frequencies (Ding et al., 2024; Lee & Yang, 2023). While decomposition improves accuracy, many studies treat components as purely mathematical constructs

and offer limited guidance on how to assign heterogeneous predictors to each component.

2.2. Textual signals and sentiment

News and online signals can precede price changes, and multiple studies show that text features complement numerical inputs. Li et al. (2019) combine convolutional neural network (CNN)-extracted features with latent Dirichlet allocation (LDA)-based topics from online news, and Wu et al. (2021) combine news and search data with VMD to improve prediction accuracy. Fang et al. (2023b) incorporate FinBERT-based sentiment representations with VMD and bidirectional gated recurrent unit (BiGRU) models to capture richer financial context. Despite these advances, many pipelines still rely on short headlines or polarity scores that can miss event structure and context, motivating richer text representations.

2.3. Multimodal fusion strategies

Multimodal learning in finance has shown that jointly modeling text and market data can outperform unimodal baselines (Hangloo & Arora, 2025). Examples include exchange rate forecasting with Bidirectional Encoder Representations from Transformers (BERT)-based sentiment and market indicators (Windsor & Cao, 2022) and multimodal fusion for Bitcoin price prediction (Han et al., 2025). In crude oil forecasting, text and numerical data are often combined but typically without a systematic comparison of fusion strategies. Moreover, practical decision-making in procurement and planning can require longer horizons than the short-term focus of many studies (Chen et al., 2015). These gaps motivate our decomposition-based multimodal framework and our explicit evaluation of early, intermediate, and late fusion for multi-week forecasting. For readability, Table 1 uses standard abbreviations, including generalized pattern matching based on genetic algorithm (GPMGA), complete ensemble empirical mode decomposition (CEEMD), support vector machine (SVM), particle swarm optimization (PSO), multi-step (MS), generalized autoregressive conditional heteroskedasticity (GARCH), mixed data sampling (MIDAS), complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN), bidirectional long short-term memory (BiLSTM), regression trees (RT) as used in AdaBoost.RT, and semantic-enhanced nonnegative matrix factorization (SeaNMF). Table 1 provides a taxonomy of crude oil forecasting studies, highlighting text transformations, numeric inputs, decomposition, fusion choices, and evaluation scope.

3. Methodology

This study develops a decomposition-based multimodal forecasting framework that integrates large-scale numerical indicators with enriched textual representations to predict multi-horizon WTI crude oil prices. The design choices are guided by two considerations. First, crude oil prices are highly non-stationary and exhibit distinct low- and high-frequency dynamics that are not well captured by a single monolithic model. Second, market-relevant information is distributed across heterogeneous sources—structured macroeconomic indicators and unstructured financial news—whose interactions are economically meaningful and should be learned jointly rather than appended post hoc. Fig. 1 summarizes the pipeline.

3.1. Data collection and feature generation

3.1.1. Data sources

The empirical analysis uses daily data from January 2015 to December 2019, a period chosen to reflect relatively stable market conditions prior to the structural disruptions induced by the COVID-19 pandemic. This design choice trades off generality for controlled evaluation. WTI spot prices serve as the forecasting target. Numerical predictors consist of more than 9,000 macroeconomic and commodity-related

¹ In this work, GPT-4.1, the latest OpenAI model available as of May 2025, is used to generate enhanced textual features from full article content rather than headlines. We set the temperature to 0 to reduce randomness and ensure consistent outputs.

Table 1
Comparison of recent crude oil price forecasting models.

Literature	Model / Type	Textual information	Transfor- mation	Numeric Features	/ Other	Decomposition	Fusion Type	Forecasting Scope	Key Contributions / Findings
Numerical / decomposition models									
Fan et al. (Fan et al., 2008)	GPMGA pattern matching		None	Prices		None	Unimodal	Multi-step	Generalized pattern matching for WTI/Brent forecasts.
Yu et al. (Yu et al., 2015)	Decomposition-ensemble		None	Prices		Custom decomposition	Ensemble	Multi-step	Data-stage reconstruction improves accuracy.
Tang et al. (Tang et al., 2020)	Multi-factor search + MEMD		Search engine data	Prices		MEMD	Ensemble	Multi-step	Multi-scale search data + MEMD enhance level/direction forecasts.
Abdollahi (Abdollahi, 2020)	CEEMD–SVM–PSO–MS–GARCH		None	Prices		CEEMD	Hybrid	Short/Med.	CEEMD + volatility modeling improves baseline performance.
Huang and Deng (Huang & Deng, 2021)	VMD–LSTM		None	Prices		VMD	Unimodal	Short/Med.	VMD + LSTM improves base models.
Fang et al. (Fang et al., 2023a)	EMD–ISBM–FNN		None	Prices		EMD	Unimodal	Multi-step	EMD variants outperform standard deep methods.
Ding et al. (Ding et al., 2024)	MIDAS + STL		None	Prices		STL	Unimodal	Multi-step	STL emphasizes trend/season/residual differences.
Zhu et al. (Zhu et al., 2025)	CEEMDAN–VMD + CNN–BiLSTM		None	Prices + macro		CEEMDAN + VMD	Ensemble	Multi-week	Dual decomposition + deep learning reports competitive MAPE and R ² values.
Text-heavy and hybrid multimodal studies									
Li et al. (Li et al., 2019)	CNN + LDA topics		CNN-extracted features + LDA-based topics	Prices		None	Sequential	Short	Text-CNN + LDA enhances oil forecasting.
Li et al. (Li et al., 2021)	VMD + BiLSTM		Sentiment index	Prices		VMD	Unimodal	Multi-step	Sentiment index captures shocks and improves forecasts.
Wu et al. (Wu et al., 2021)	CNN + VMD		News + search	Prices		VMD	Hybrid	Multi-step	Text + trends reduce MAPE.
Bai et al. (Bai et al., 2022)	AdaBoost.RT + SeaNMF		SeaNMF topical sentiment	None		None	Early	Short	Headline semantic features aid prediction.
Fang et al. (Fang et al., 2023b)	FinBERT + VMD + attention-based Bi-GRU		FinBERT sentiment	Prices		VMD	Early fusion	Multi-step	FinBERT + VMD + attention improves accuracy.
Kaplan et al. (Kaplan et al., 2026)	CrudeBERT sentiment		Domain-tuned language model (LM)	Prices		None	Unimodal	Multi-step	Crude-specific BERT enhances sentiment signals.
This study	STL + LSTM		LLM summaries + FinBERT	Prices + macro + text		STL	Early / Intermediate / Late	Multi-week + directional	Generative + semantic features; systematic fusion comparison; interpretable decomposition.

indicators collected from the U.S. Energy Information Administration (EIA), the Organisation for Economic Co-operation and Development statistics portal (OECD-STAT), and the Federal Reserve Economic Data (FRED) databases. These variables span production, inventory, trade flows, prices of related energy products, and broad macroeconomic conditions, including other liquids (propane and butane) and consumer price index (CPI).

Textual information is drawn from financial news articles published in the *Commodities* section of Investing.com. Rather than relying on short headlines, the full article content is retained to preserve contextual detail that may precede or explain price movements.

3.1.2. LLM-enhanced text features

To mitigate the limitations of short and ambiguous headlines, this study uses GPT-4.1, the latest OpenAI LLM model available as of May

2025, to generate enhanced textual features from full article content rather than headlines. The input explicitly excludes original headlines to prevent redundancy, and the model is configured with a temperature of 0 to reduce randomness. The full prompt templates and feature definitions are provided in [Appendix A](#).

The generated summaries are subsequently encoded using FinBERT, a language model pre-trained on financial corpora. Token-level embeddings are aggregated via masked mean pooling to obtain a fixed-length sentence embedding of dimension 768 for each weekly document. Compared with polarity scores or keyword counts, this representation preserves contextual meaning and allows downstream models to learn non-linear interactions between narrative information and numerical indicators.

We note that GPT-4.1's training corpus extends through and beyond the 2015–2019 evaluation window, which raises the question of

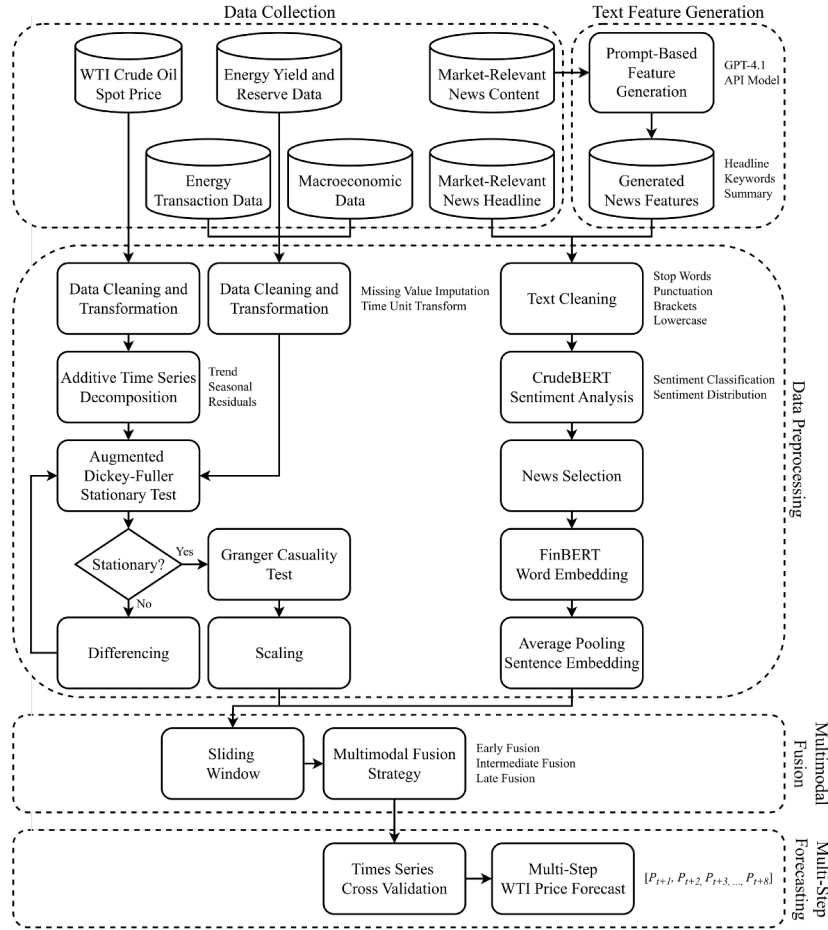


Fig. 1. Framework of the proposed multimodal fusion pipeline.

whether its text transformations carry implicit post-hoc knowledge. In this pipeline the model functions as a constrained text extractor that receives only the article content, produces a headline, five keywords, and a one-line summary of what the article states, and operates at temperature zero to suppress creative elaboration. The information it returns is therefore bounded by what the article itself contained, an analogous constraint to that faced by any knowledgeable analyst summarizing the same historical text. A residual concern is that the model's word choices may subtly reflect post-hoc framing, but this is a stylistic second-order effect and does not affect the numerical inputs or training targets. FinBERT's corpus, which consists predominantly of financial filings and news predating 2020, introduces no analogous concern.

3.2. Data preprocessing

3.2.1. Numerical preprocessing and decomposition

All numerical time-series variables were aligned to a daily frequency to ensure temporal consistency across heterogeneous data sources. Missing observations, which primarily arise from weekends and public holidays, were handled using linear interpolation so as to preserve continuity without introducing artificial jumps into the series.

Most gaps arise from non-trading days and asynchronous release schedules across numerical and textual sources. Linear interpolation is applied only to bridge structural calendar gaps and preserve temporal alignment before weekly aggregation; it is not used to infer unobserved weekend or holiday market movements. Because the forecasting task is evaluated at weekly horizons, short-run interpolation noise is further attenuated after aggregation and is unlikely to introduce systematic bias.

To enhance interpretability and facilitate component-wise modeling, the WTI crude oil price series was decomposed using STL decomposition, which has recently been shown to provide robust and economically interpretable components in forecasting applications (Ding et al., 2024). Although EMD and VMD are widely used in financial forecasting, their frequency-based modes are harder to interpret economically. STL yields trend, seasonal, and residual components with direct economic meaning: the trend captures long-run macroeconomic and structural market forces, the seasonal component reflects recurring demand and operational cycles, and the residual captures irregular shocks often associated with news and sentiment. This interpretability supports the component-specific multimodal design adopted in this study.

STL decomposes the observed price into additive components:

$$P_t = T_t + S_t + R_t, \quad (1)$$

where P_t denotes the observed price at time t , and T_t , S_t , and R_t represent the trend, seasonal, and residual components, respectively. This decomposition remains stable under structural variation while preserving clear economic interpretation at the component level. The stationarity of all decomposed components and candidate numerical predictors was examined using unit-root testing based on the Augmented Dickey-Fuller framework, which remains a standard diagnostic tool in applied time-series analysis. Non-stationary series, including the trend component, were transformed via first-order differencing until stationarity was achieved. This ensures that all variables entering downstream statistical procedures share a common order of integration.

Given the substantial dimensionality of the numerical feature pool (nearly 9,000 macroeconomic and commodity indicators), predictive screening was conducted using Granger-causality-based tests, which are

commonly adopted in high-dimensional economic forecasting to identify variables with incremental predictive content. Only variables exhibiting statistically significant predictive associations with the corresponding price components were retained. Applying Granger tests across nearly 9,000 candidates simultaneously raises familywise error concerns under a nominal threshold. Here, however, the screening step serves as dimensionality reduction, eliminating variables with no predictive association before model training, rather than as a formal causal inference procedure. Spuriously retained variables compete against genuinely predictive features during optimization; dropout and early stopping attenuate their influence, and the expanding-window cross-validation provides the honest test of out-of-sample generalization. The consistency of results across three error metrics and all eight forecast horizons supports the view that the retained feature set is not materially distorted by selection noise. Finally, all selected numerical features were standardized using z-score normalization to stabilize neural network training.

The macroeconomic candidate pool is high dimensional by construction, so overfitting control is built into both preprocessing and model training. Granger-causality screening reduces the initial set of nearly 9,000 variables to 430 predictors with component-specific predictive content, substantially lowering input noise before learning begins. Decomposition further constrains each network to a narrower signal subspace by training separate models for trend, seasonal, and residual components. During optimization, dropout, early stopping, and validation-based hyperparameter selection regularize model capacity, and the expanding-window validation protocol repeatedly evaluates unseen time segments to verify that performance gains persist out of sample.

3.2.2. Textual data preprocessing

The textual preprocessing pipeline was designed with the objective of retaining semantically rich and economically meaningful information while minimizing noise commonly present in raw financial news. Full-length news articles were first summarized using an LLM, producing concise, context-aware summaries that preserve key events and narratives. Standard text cleaning procedures were applied, including lower-casing, punctuation removal, and stop-word filtering.

To ensure that textual inputs were closely aligned with oil price dynamics, a two-stage language model framework was adopted. In the first stage, we applied Crude-BERT (Kaplan et al., 2026), a domain-specific transformer model trained on energy-market news, to filter daily articles and assess their relevance to oil price movements. For each day, the dominant sentiment polarity was determined by majority voting, and the single article with the highest classification confidence was selected as the representative news item.

Second, weekly textual representations were constructed by concatenating the selected daily summaries. These weekly documents were embedded using FinBERT (Liu et al., 2021), a transformer model pretrained on large-scale financial corpora. Masked mean pooling was applied to token-level embeddings, producing a fixed 768-dimensional vector for each week. This representation captures deep contextual and semantic information beyond simple sentiment scores.

3.3. Multimodal fusion and forecasting architecture

Building on the decomposed price representation, the forecasting framework models each STL component (trend, seasonal, and residual) independently using recurrent neural networks with LSTM units, which are well suited for capturing temporal dependencies in economic time series (Yu et al., 2019). This architectural choice reflects the distinct statistical characteristics and underlying economic drivers associated with each component.

- **Trend model:** Incorporates lagged trend differences, selected macroeconomic indicators, and textual embeddings to capture long-term structural dynamics.

- **Seasonal model:** Uses only lagged seasonal values, reflecting the self-driven and repetitive nature of seasonality.
- **Residual model:** Combines lagged residuals with textual embeddings to model event-driven shocks and short-term irregularities.

For the trend and residual components, we systematically examined alternative multimodal fusion strategies that have been extensively discussed in recent multimodal learning surveys (Hangloo & Arora, 2025).

1. **Early fusion (EF):** Numerical and textual features are concatenated at the input layer and processed by a single LSTM (Fig. 2).
2. **Intermediate fusion (IF):** Each modality is encoded by a separate LSTM branch, and their latent representations are fused prior to prediction (Fig. 3).
3. **Late fusion (LF):** Separate LSTM models are trained for each modality, and their predictions are combined via a final dense layer (Fig. 4).

Forecasting uses a sliding-window design: the previous eight weeks are used to predict the next eight weeks, consistent with procurement lead times. Inputs are aligned Monday-to-Monday to match weekly planning cycles used in practice. Models are evaluated with expanding-window cross-validation under strict chronological ordering. In each fold, data are split into 70% training, 15% validation, and 15% testing, and the training window expands in later folds as new observations become available. Section IV reports fold-wise means and standard deviations, quantifying both average accuracy and stability across evaluation windows while limiting information leakage. We note that Granger-causality screening was conducted on the complete 2015–2019 sample prior to fold construction rather than re-run within each training window, introducing a selection-stage dependency on observations that appear in later test folds. Four considerations bound the practical implications. First, fold-internal screening on early expanding windows would test nearly 9,000 candidates against training histories of roughly 100 weekly observations, a regime in which Granger test statistics are unreliable and false-discovery rates are elevated; global screening on the full five-year sample uses the maximum available statistical power to identify stable predictive associations, making it preferable to fold-specific screening in early folds on statistical grounds alone. Second, the pre-pandemic period is structurally stable, so the predictive associations identified on the full series are unlikely to differ materially from those that fold-specific screening on any sufficiently long training window would have selected. Third, the leakage operates only at the variable selection stage; all model parameters are estimated exclusively within each fold's training data. Fourth, if selection-stage look-ahead were materially distorting generalization, performance would be expected to deteriorate as test windows advance; this pattern is not observed in the fold-wise results reported in Section 4.

3.4. Hybrid loss function

To jointly account for predictive accuracy in both magnitude and direction, a hybrid loss function was adopted to better align the learning objective with decision-oriented forecasting requirements:

$$L_{\text{hybrid}} = \lambda_1 L_{\text{base}} + (1 - \lambda_1) L_{\text{dir}}, \quad (2)$$

where L_{base} denotes a standard regression loss and L_{dir} penalizes incorrect directional predictions.

For the residual component, MSE was used as L_{base} without a directional term, reflecting its noisy nature. For the trend and seasonal components, a robust regression loss based on the Huber formulation was employed to mitigate the influence of extreme observations, while an additional directional penalty was introduced to discourage sign errors in predicted price movements. The weighting parameter λ_1 was selected via validation.

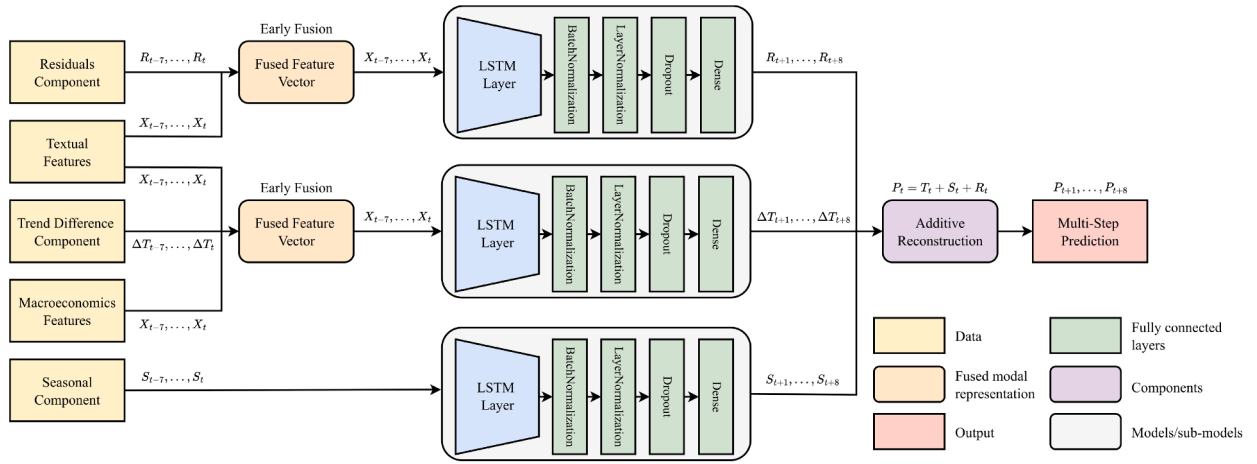


Fig. 2. Early fusion architecture for multimodal time-series forecasting.

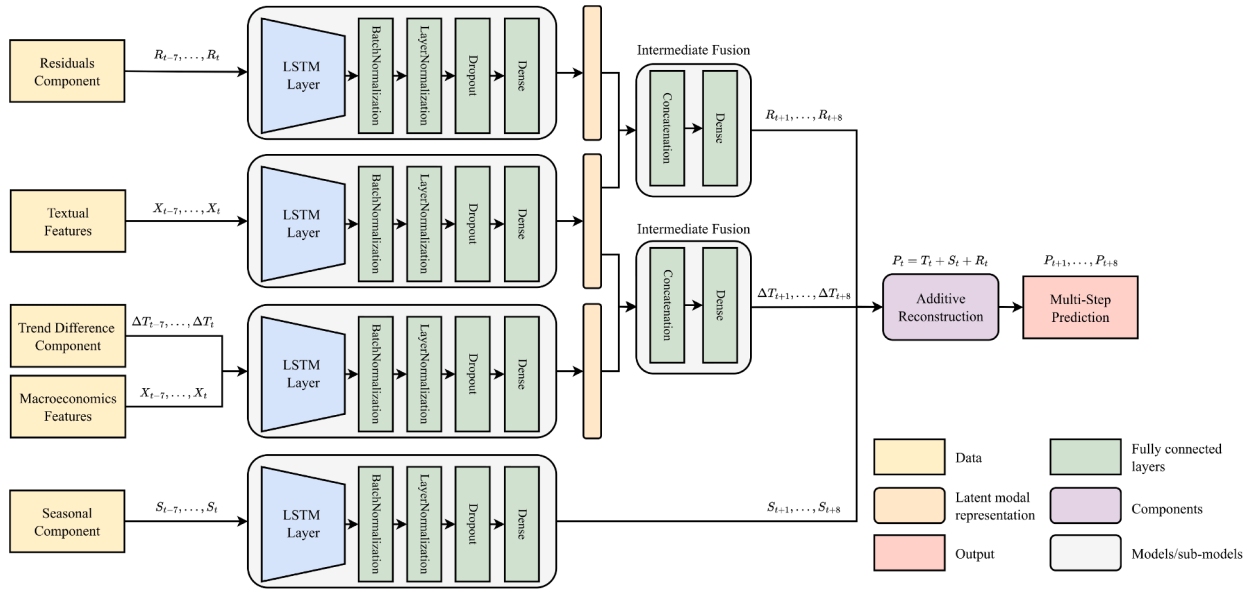


Fig. 3. Intermediate fusion architecture for multimodal time-series forecasting.

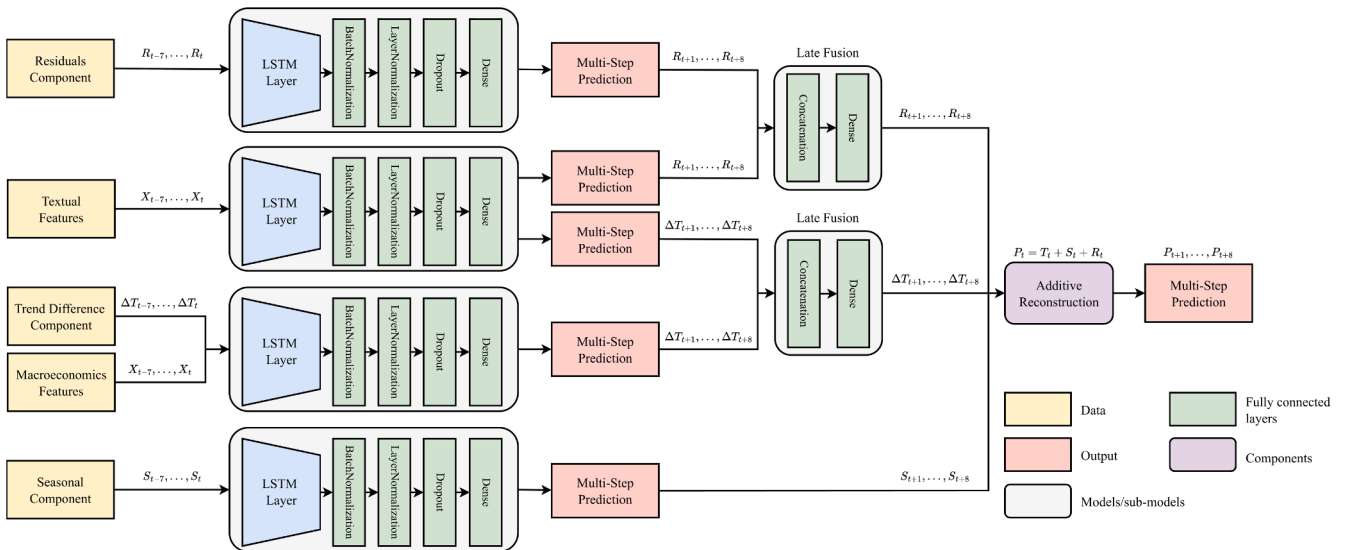


Fig. 4. Late fusion architecture for multimodal time-series forecasting.

The directional loss is defined as

$$L_{\text{dir}} = \frac{1}{n} \sum_{i=1}^n p \cdot \mathbb{I}[s_i \neq \hat{s}_i] \cdot |y_i - \hat{y}_i|, \quad (3)$$

where n is the number of samples in the batch, s_i and $\hat{s}_i$ denote the thresholded signs of actual and predicted price changes, respectively, and p is a penalty factor.

4. Empirical results

The empirical study evaluates whether event-aware textual signals, when fused with large-scale numerical indicators under an STL–LSTM decomposition, improve multi-horizon WTI forecasting. We report point-forecast errors over an 8-week horizon (Weeks 1–8) using MAPE, MSE, and SMAPE, and we additionally assess directional correctness via a trend-accuracy score defined by a “flat” band.

4.1. Fusion strategy implementations

We evaluate three integration strategies that differ in when modalities are combined. Early fusion concatenates lagged prices, macro indicators, and FinBERT embeddings at the input and feeds them to a shared LSTM, enabling rich cross-modal dependencies but reducing interpretability and increasing dimensionality (and thus overfitting risk when signals are noisy). Intermediate fusion encodes each modality in parallel LSTM branches and merges their latent states, preserving modality structure while allowing interaction; this balance comes with higher complexity and more tuning of the merge layer. Late fusion keeps each modality independent and aggregates only final predictions, improving robustness when modality quality varies but limiting the model’s ability to learn deep cross-modal representations.

4.2. Analysis of selected features and decomposition

The Granger causality screening retains 430 numerical variables as statistically significant predictors for the trend component. The top 15 features (Fig. 5) are dominated by energy-market indicators (e.g., WTI/Brent/Dubai futures and refined product prices) and major macroeconomic indices such as CPI and producer price index (PPI), aligning with the economic channels through which demand, inflation, and benchmark spreads transmit into crude price formation.

The STL decomposition provides an interpretable separation into a long-run trend, a recurring 28-day seasonal pattern tied to monthly inventory updates, contract renewals, and shipping schedules, and a high-frequency residual. This structure matters empirically because it supports component-wise assignment of modalities: persistent numerical drivers are naturally aligned with the trend/seasonal parts, while event-rich text is most plausibly informative for the residual shocks. Unlike VMD or ensemble empirical mode decomposition (EEMD) used in prior studies (e.g., Huang and Deng (2021)), we select STL to preserve economic interpretability, mapping macroeconomic indicators to the trend component and textual shock signals to the residual. (The decomposition plots are provided in Fig. 6.)

The decomposition design also provides a concrete mechanism for multimodal gains. Structural supply-demand conditions evolve gradually and are therefore concentrated in the trend component, where macroeconomic predictors are most informative. Event-driven disturbances, including geopolitical announcements and inventory surprises reported in financial news, enter primarily through the residual component. Assigning modalities to components on this economic basis reduces cross-component noise and improves the model’s ability to extract cross-modal signal where interaction is most likely.

4.3. Comparison of modalities and fusion strategies

Tables 2, 3, and 4 report error metrics across the 8-week horizon for all modalities, including baselines and fusion strategies. The results indi-

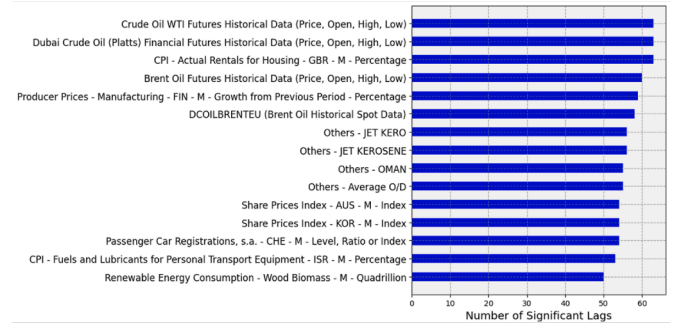


Fig. 5. Top 15 most influential variables for oil price trend forecast based on Granger causality.

cate that multimodal fusion models consistently outperform all baselines in this evaluation setting. The Early Fusion model reports the lowest error metrics across all eight weeks. For instance, in Week 8 it achieves a MAPE of 5.78%, a notable improvement over the strongest baseline (Crude Oil + Numerical Features, 6.39%).

Economically, the early-fusion advantage implies that crude oil prices respond to interacting rather than separable signals. Macroeconomic variables provide the structural backdrop, while news narratives reflect emerging events that can strengthen or offset that backdrop. By learning a shared representation at the input stage, early fusion captures these interactions directly, including cases where narrative shocks reprice macro-driven expectations. This matters most at longer horizons, where structural forces and expectation updates jointly shape price paths.

The Intermediate Fusion model is the second-best performer. While its error metrics are slightly higher than early fusion, it exhibits the lowest standard deviation in the first three weeks, suggesting high stability for short-term forecasting. The Late Fusion model performs weakest among the fusion strategies, highlighting the importance of learning interactions between modalities during the training process.

Table 5 reports trend accuracy and highlights the advantage of structured numerical features. This suggests that while textual features are important for reducing overall prediction error (magnitude), the fundamental direction of price trends is more strongly dictated by macroeconomic and market-related numerical data.

4.4. Visual analysis of forecasts

Fig. 7 compares the forecasts of the three fusion strategies against the best-performing baseline model (Crude Oil + Numerical Features) for Weeks 1 to 8. The plots confirm that the Early Fusion model tracks the actual price series most closely, especially during periods of high volatility. Intermediate fusion exhibits smoother trajectories and can lag abrupt movements, whereas late fusion tends to underreact.

4.5. Stability across validation folds (Weeks 1–8)

Following the thesis evaluation protocol, model comparisons are summarized by fold-wise means and standard deviations of errors. Tables 2, 3, and 4 show that Early Fusion achieves the lowest average error across horizons while maintaining strong stability.

For example, in MAPE, Early Fusion records a standard deviation of 0.1985 at Week 1 and remains more stable than both Intermediate and Late Fusion by Week 8 (1.9095 vs. 2.4942 and 2.3925, respectively). A similar long-horizon pattern appears for MSE and SMAPE, where Early Fusion shows lower dispersion than both alternative fusion strategies from mid-to-long horizons. Lower dispersion across folds indicates more stable performance across evaluation windows.

Table 2
MSE-based performance comparison across modalities.

Wk.	Crude	Oth. Num.	Text	Crude + Text	Crude + Num.	Early	Interm.	Late
1	2.0612 (0.5566)	1.6882 (0.3851)	2.2821 (0.7982)	1.5624 (0.3721)	1.5728 (0.4372)	1.3662 (0.2567)	1.4397 (0.1526)	1.5641 (0.2019)
2	5.3028 (1.9360)	2.9818 (0.8080)	5.9077 (2.5907)	3.8577 (1.3344)	2.5843 (0.7158)	2.2646 (0.6597)	2.5765 (0.3750)	2.9151 (0.8489)
3	11.5738 (4.6053)	5.3170 (1.5207)	12.6887 (5.5979)	9.2715 (3.8292)	5.0703 (1.6004)	4.0811 (1.3251)	4.5275 (0.7794)	5.1345 (1.6195)
4	18.4875 (8.0592)	8.2224 (2.0622)	19.9937 (9.7844)	16.3391 (7.4558)	7.7103 (2.3528)	6.8115 (2.1877)	7.2117 (1.9914)	8.2396 (2.9030)
5	28.0968 (12.7538)	13.4948 (4.7568)	29.4645 (16.2086)	25.6754 (13.3041)	13.0787 (5.5096)	12.1167 (4.1450)	12.7934 (5.2442)	14.1733 (5.9476)
6	35.5116 (16.5181)	17.8362 (8.6114)	36.3603 (22.4380)	33.0305 (19.6308)	17.0468 (9.2389)	15.6816 (6.5948)	17.3495 (10.0302)	19.4247 (10.7646)
7	41.3227 (19.9130)	22.6385 (14.2749)	41.2812 (28.2740)	38.8072 (24.9415)	22.3495 (14.7814)	19.3489 (10.6354)	21.8193 (14.5785)	24.5525 (16.2069)
8	48.2662 (24.9806)	29.5331 (20.4780)	47.5564 (35.7391)	44.6239 (30.3780)	28.2956 (19.9613)	23.5886 (15.7357)	27.6517 (21.5045)	30.7402 (22.8446)

Note: Lower is better. Standard deviation in parentheses.

Table 3
MAPE-based performance comparison across modalities (%).

Week	Crude	Other Num.	Textual	Crude + Text	Crude + Num.	Early	Interm.	Late
1	1.9304 (0.3435)	1.7193 (0.2101)	2.0136 (0.4660)	1.6526 (0.2516)	1.6171 (0.2694)	1.5366 (0.1985)	1.5825 (0.1658)	1.6341 (0.1320)
2	3.1426 (0.6333)	2.2686 (0.3832)	3.2792 (0.7888)	2.6094 (0.5230)	2.1008 (0.3512)	1.9248 (0.2538)	2.0613 (0.1195)	2.2158 (0.2658)
3	4.6904 (0.9097)	3.0913 (0.5817)	4.9009 (1.0321)	4.1201 (0.8715)	2.9826 (0.5211)	2.6298 (0.4597)	2.7643 (0.3184)	2.9728 (0.4670)
4	5.8364 (1.2925)	3.8195 (0.6218)	6.0532 (1.4313)	5.3951 (1.0653)	3.6439 (0.7026)	3.4484 (0.6543)	3.5601 (0.6578)	3.7991 (0.7410)
5	7.1096 (1.7083)	4.7280 (0.9166)	7.2175 (1.9768)	6.6637 (1.4727)	4.5437 (1.0012)	4.5131 (0.8579)	4.5536 (1.0907)	4.8848 (1.0755)
6	7.9050 (1.9962)	5.2722 (1.1671)	7.8578 (2.4195)	7.3963 (1.8760)	5.0391 (1.2786)	4.9836 (1.0343)	5.0587 (1.5891)	5.4521 (1.5114)
7	8.5014 (2.2729)	5.6969 (1.6050)	8.2194 (2.8261)	7.8808 (2.2564)	5.6675 (1.7637)	5.3479 (1.3865)	5.4469 (1.9413)	5.9152 (1.8762)
8	9.2093 (2.7094)	6.4458 (2.1094)	8.7389 (3.3754)	8.3441 (2.7153)	6.3927 (2.2037)	5.7784 (1.9095)	6.0519 (2.4942)	6.5416 (2.3925)

Note: Lower is better. Standard deviation in parentheses.

Table 4
SMAPE-based performance comparison across modalities (%).

Week	Crude	Other Num.	Textual	Crude + Text	Crude + Num.	Early	Interm.	Late
1	1.9253 (0.3350)	1.7130 (0.2116)	2.0006 (0.4536)	1.6433 (0.2422)	1.6121 (0.2704)	1.5339 (0.1998)	1.5770 (0.1699)	1.6290 (0.1353)
2	3.1274 (0.5985)	2.2608 (0.3790)	3.2452 (0.7422)	2.5832 (0.4927)	2.0928 (0.3454)	1.9204 (0.2507)	2.0560 (0.1181)	2.2073 (0.2542)
3	4.6619 (0.8231)	3.0784 (0.5690)	4.8418 (0.9272)	4.0579 (0.7993)	2.9690 (0.5006)	2.6179 (0.4506)	2.7484 (0.3135)	2.9563 (0.4422)
4	5.7755 (1.1356)	3.7844 (0.5721)	5.9435 (1.2437)	5.2719 (0.9117)	3.6005 (0.6388)	3.4029 (0.6150)	3.5187 (0.6232)	3.7534 (0.6926)
5	7.0170 (1.4851)	4.6406 (0.8197)	7.0288 (1.6853)	6.4601 (1.2234)	4.4396 (0.8991)	4.4170 (0.7786)	4.4570 (0.9971)	4.7891 (0.9742)
6	7.7882 (1.7268)	5.1243 (0.9921)	7.5918 (2.0079)	7.1107 (1.5162)	4.8820 (1.1082)	4.8430 (0.9013)	4.8870 (1.4287)	5.2772 (1.3133)
7	8.3585 (1.9777)	5.4644 (1.3343)	7.8760 (2.3000)	7.5143 (1.8014)	5.4321 (1.4926)	5.1444 (1.1858)	5.2123 (1.6979)	5.6581 (1.5739)
8	9.0520 (2.4275)	6.1450 (1.7289)	8.3189 (2.7411)	7.8959 (2.1682)	6.0958 (1.8454)	5.5080 (1.6318)	5.7400 (2.1293)	6.2043 (1.9668)

Note: Lower is better. Standard deviation in parentheses.

4.6. Comparison with recent models

To position the proposed framework, we compare our best model (Early Fusion) with several recent crude oil price forecasting models in Table 6. We acknowledge that a direct, one-to-one comparison is challenging due to variations in datasets, forecast horizons, and evaluation periods. Nonetheless, this benchmark provides an evidence-based perspective on our model's performance relative to advanced methods.

The comparison highlights two practical patterns. First, at a comparable monthly decision scale, our 4-week result (MAPE 3.45%, MSE 6.81, trend accuracy 81.51%) has numerically lower MAPE than Zhang et al.'s 1-month MAPE of 4.11% and also higher than their reported directional accuracy of 74.29%. Second, relative to short-horizon daily models such as Zhu et al. (1-day MAPE 3.66%), our framework maintains competitive error while extending the forecast horizon to operationally harder multi-week settings. Within our own rows, error grows

Table 5
Trend accuracy ($\pm 1\%$ threshold, %) across different modalities.

Week	Crude	Other Num.	Textual	Crude + Text	Crude + Num.	Early	Interm.	Late
1	66.41 (2.53)	69.14 (4.82)	64.58 (1.44)	72.01 (4.35)	71.35 (5.55)	70.18 (3.94)	71.74 (4.97)	71.35 (4.33)
2	64.97 (3.63)	77.34 (5.85)	64.71 (6.10)	74.74 (6.22)	77.86 (6.70)	79.17 (4.80)	77.73 (6.37)	77.34 (4.72)
3	59.24 (8.49)	74.87 (4.80)	55.73 (8.83)	67.06 (4.33)	76.69 (4.65)	81.51 (3.70)	78.26 (4.65)	78.26 (4.63)
4	52.47 (8.11)	78.78 (5.07)	50.13 (8.28)	63.02 (2.71)	82.29 (1.64)	81.51 (3.81)	80.99 (3.58)	78.78 (0.80)
5	49.22 (8.02)	71.74 (5.67)	44.66 (6.59)	58.07 (2.08)	76.17 (1.69)	73.44 (6.11)	78.12 (2.83)	70.96 (4.61)
6	49.87 (10.41)	72.53 (4.73)	46.35 (5.73)	54.69 (2.49)	73.57 (3.32)	71.48 (4.72)	74.09 (2.12)	69.53 (4.47)
7	47.53 (14.34)	68.75 (5.42)	48.05 (2.83)	54.04 (2.12)	67.58 (4.15)	69.79 (4.63)	70.31 (6.66)	66.93 (4.47)
8	46.09 (16.40)	69.40 (4.33)	50.39 (2.09)	57.42 (2.61)	70.05 (6.62)	72.01 (6.24)	74.48 (6.41)	69.66 (2.89)

Note: Higher is better. Standard deviation in parentheses.

Table 6
Comparison of crude oil price forecasting metrics across recent works.

Reference	Model/Methodology	Horizon	RMSE	MAE	MSE	MAPE (%)	Trend (%)
Zhu et al. (2025) (Zhu et al., 2025)	CEEMDAN-VMD-CNN-BiLSTM	Daily	3.9537	2.8250	15.6319	3.6648	–
Zhang et al. (2024) (Zhang et al., 2024)	Attention-PCA + ABC Fusion	Monthly	3.7221	3.0249	–	4.1119	74.29
Fang et al. (2023a) (Fang et al., 2023a)	EMD-ISBM-FNN (Iterative)	Daily	1.3429	0.9731	–	1.5900	–
This study (Early Fusion)	STL-LSTM + LLM Text	1-Week	1.1697	–	1.3662	1.5366	70.18
This study (Early Fusion)	STL-LSTM + LLM Text	4-Week	2.6099	–	6.8115	3.4484	81.51
This study (Early Fusion)	STL-LSTM + LLM Text	8-Week	4.8568	–	23.5886	5.7784	72.01

gradually from Week 1 to Week 8 (MAPE 1.54% to 5.78%), but directional performance remains consistently usable (70.18% to 81.51% across reported horizons), suggesting that the model preserves signal quality even as horizon uncertainty increases.

Overall, the combined evidence from absolute error and directional metrics indicates that integrating large-scale numerical indicators with LLM-enhanced text via early fusion provides a better accuracy–stability tradeoff than competing strategies for multi-horizon crude oil forecasting.

5. Conclusion and future work

5.1. Conclusion

This study develops a decomposition-based multimodal forecasting framework for crude oil prices that explicitly addresses two persistent weaknesses in the literature: the shallow use of textual information and the opaque role of signal decomposition in deep forecasting models. By integrating LLM-enhanced news representations with a large set of structured macroeconomic and commodity indicators, the proposed framework demonstrates that textual context and numerical fundamentals are complementary rather than substitutable sources of predictive information. The empirical results indicate that, within this experimental setting, fusing these modalities at an early stage yields improved accuracy and stability across multi-week horizons, particularly beyond short-term forecasts.

A central contribution lies in the use of STL decomposition to impose an economically interpretable structure on the forecasting task. Separating prices into trend, seasonal, and residual components allows different data modalities to be assigned where they are most informative. The results indicate that numerical indicators dominate long-run directional signals, while LLM-enhanced textual features provide critical context for explaining the magnitude and timing of deviations, especially in the trend and residual components. This component-wise assignment not only improves forecasting performance but also enhances interpretability,

making the model more suitable for decision support in procurement and risk management settings.

The risk-management utility targeted here is operational and procurement risk, covering advance purchasing decisions, inventory rebalancing, contract timing, and short-term hedging, all exposed to weekly price dynamics within a functioning market. This is categorically distinct from crisis-period tail risk, such as the demand collapse and market microstructure failures associated with COVID-19, which constitute structural breaks that no text-based forecasting model can address regardless of the evaluation window; meeting that challenge requires dedicated regime-switching or structural break detection methods that lie outside the scope of this study.

The systematic comparison of early, intermediate, and late fusion strategies further clarifies a question that is often left implicit in prior work. Across all evaluated horizons and error metrics, early fusion consistently outperforms competing architectures, with lower mean errors and lower or comparable standard deviations across validation folds. These findings suggest that allowing cross-modal interactions to be learned directly at the representation level is crucial when dealing with high-dimensional numerical inputs and dense semantic text embeddings.

Overall, the proposed framework advances crude oil price forecasting along three dimensions: richer textual feature construction through LLMs, interpretable decomposition aligned with economic intuition, and principled multimodal fusion validated by fold-wise mean and stability analysis. The resulting model reports competitive multi-horizon forecasting performance while remaining transparent enough to support practical decision-making. Each STL component targets a distinct planning problem. Trend forecasts, governed by macroeconomic indicators, inform the recurring contract mode decision; a manager anticipating rising structural prices over a four-to-eight-week horizon can commit to fixed-price supply before costs increase, while a falling trend supports deferred commitment or floating exposure. Seasonal forecasts, isolating a predictable 28-day demand cycle, support inventory timing decisions, particularly buffer stock adjustments around monthly delivery and contract renewal windows.

The residual component, where textual embeddings are most informative, captures event-driven deviations that call for rapid hedging responses rather than structural repositioning. At the four-week horizon, 81.51% directional accuracy provides a concrete operational benchmark. The model correctly identifies the favorable direction on the binary fixed-versus-floating contract decision more than four out of five times across the evaluation period.

5.2. Future work

Several directions arise from this study. First, the current study focuses on a pre-pandemic sample to avoid structural breaks associated with extreme events. Extending the framework to explicitly handle regime shifts, such as those induced by geopolitical conflicts or global crises, would improve its robustness in more volatile environments. Incorporating regime-switching mechanisms or adaptive reweighting of modalities may allow the model to adjust its reliance on textual versus numerical information over time. The focus of the current study is operational risk under functioning market conditions; crisis-period structural breaks such as pandemic-induced demand collapse represent a categorically different problem class requiring architectural extensions beyond text-based forecasting.

Second, while this study employs LSTM-based architectures for temporal modeling, recent advances in transformer-based time-series models offer promising alternatives for capturing long-range dependencies. Future research could examine whether the observed advantage of early fusion persists under attention-based architectures and whether cross-modal attention mechanisms further enhance interpretability.

Third, the generative LLM component is currently used to summarize and structure news content prior to embedding. More tightly coupling text generation and forecasting—for example, through joint optimization or feedback-driven prompt adaptation—may yield additional gains, particularly in rapidly evolving market conditions. Evaluating the sensitivity of forecasts to prompt design and model updates also remains an open methodological question.

Finally, from an application perspective, extending the framework to other energy commodities or regional markets would help assess its generalizability. Integrating forecast outputs into downstream decision models, such as procurement optimization or hedging strategies, would further clarify the economic value of multimodal, LLM-enhanced forecasting systems.

Availability of data and material

All the data used in this study are described or available from public datasets.

Funding

This research was funded by the National Science and Technology Council of Taiwan (NSTC113-2221-E-002-175-MY3).

CRedit authorship contribution statement

Yen-Wen Chen: Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing - review & editing; **Welbey Prasadipta:** Data curation, Formal analysis, Methodology, Software, Visualization, Writing - original draft; **Chia-Yen Lee:** Conceptualization, Funding acquisition, Methodology, Resources, Supervision, Writing - review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Chia-Yen Lee reports financial support was provided by National Science and Technology Council of Taiwan. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Prompts for LLM feature generation (GPT-4.1)

Reproducibility of large language model (LLM)-based feature engineering depends more on constraining the generation problem than on model capacity. The prompts therefore enforce a narrow output format and avoid discretionary narrative. We instruct the LLM (GPT-4.1) to act as a *helpful economist and journalist* and set the temperature to 0 to minimize stochastic variation across application programming interface (API) calls.

Generated headline

A concise, high-quality headline that is attractive, clear, accurate, and inoffensive.

Prompt: Please come up with a high-quality (attractive, clear, accurate, and inoffensive) headline for the following news article; please only generate the headline without quotation marks.

Keywords

Five short keywords that capture the main themes of the article.

Prompt: Please identify five keywords for the following news article, and make each one or two words long (when generating the keywords, please use the format: ..., ..., ...).

One-line summary

A one-line summary that conveys the article's key message as concisely as possible.

Prompt: Generate a short summary of the following news article in as few words as possible; please ensure the summary is in one line without creating a new line for a new sentence.

Appendix B. LLM Token, Latency, and Cost (GPT-4.1)

Because generative preprocessing can become a hidden bottleneck, we report the token footprint and runtime behavior for the full corpus-level feature generation. We processed 9,012 news articles published between 2015 and 2019. The average input length per article was 926 ± 74 tokens. The generation tasks produced the following mean output lengths (mean $\pm$ standard deviation): generated headline 16.6 ± 6.2 tokens, keywords 17.2 ± 5.7 tokens, and one-line summary 15.1 ± 7.0 tokens.

The mean latency per API call was 0.32 ± 0.09 s. Processing the entire corpus cost 53.67 U.S. dollars (USD).

Table 7
Hyperparameter search ranges for LSTM models.

Hyperparameter	Search range
Number of LSTM layers	1, 2, 3
Hidden units per layer	16, 32, 64, 128, 256
Dropout rate	0.1, 0.2, 0.3, 0.4
Learning rate	0.01, 0.001, 0.0001
Batch size	16, 32, 64, 128

Table 8
Optimal hyperparameters for the early fusion model.

Hyperparameter	Optimal value
Number of LSTM layers	2
Hidden units per layer	128
Dropout rate	0.2
Learning rate	0.001
Batch size	32
Optimizer	Adam
Epochs (early stopping)	100 (patience = 10)

Table 9
ADF test results for WTI price and decomposed components.

Series	Original	1st diff.
WTI crude oil spot price	0.3001 (non-stationary)	< 0.001 (stationary)
Trend component	0.1118 (non-stationary)	< 0.001 (stationary)
Seasonal component	< 0.001 (stationary)	–
Residual component	< 0.001 (stationary)	–

Appendix C. Hyperparameter tuning

Hyperparameters were tuned with an explicit bias toward stable long-horizon generalization rather than squeezing marginal gains from aggressive search. We therefore used a grid search (rather than high-variance meta-heuristics) and optimized against validation MAPE for a 4-week-ahead target, consistent with the expanding-window evaluation protocol. Table 7 lists the search ranges, and Table 8 reports the selected hyperparameters.

For the hybrid loss, we tuned $\lambda_1 \in 0.60, 0.70, 0.80, 0.85, 0.90, 0.95$ and $p \in 0.5, 1.0, 2.0$ via validation within expanding-window cross-validation. The final settings were: trend (Huber + directional) $\lambda_1 = 0.85$, $p = 1.0$; seasonal (Huber + directional) $\lambda_1 = 0.90$, $p = 1.0$; residual uses MSE only.

Appendix D. Additional analyses

Table 9 reports the Augmented Dickey-Fuller (ADF) tests and complements the decomposition view in Fig. 6.

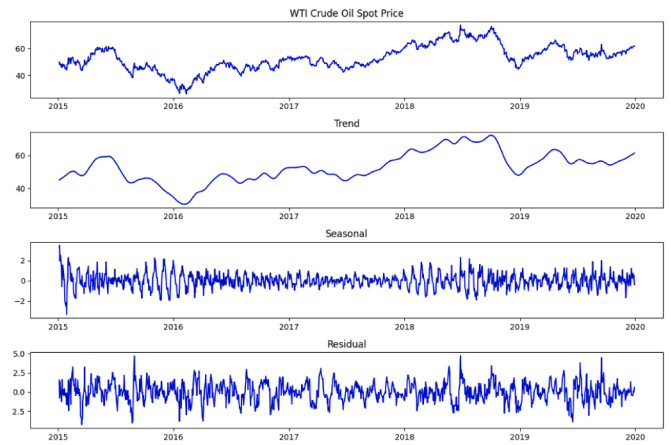


Fig. 6. Graphical representation of WTI price and its decomposed components.

Appendix E. Additional forecasting results

Fig. 7a–h compare the best baseline model against the three fusion strategies for Weeks 1–8. These visualizations complement the aggregate metrics by revealing how each fusion design behaves during transient adjustments, where error distributions are typically non-Gaussian and directionality matters for downstream decisions.

Appendix F. Forecasting results across all baseline and fusion strategies

Fig. 8a–h report the full 8-week trajectory comparisons between the realized WTI series and predictions generated by all modality combinations (five baselines plus early, intermediate, and late fusion). These plots clarify two consistent patterns: fusion tends to improve stability under volatility clustering, and modality pairing of price with structured numerical features remains competitive even when textual signals are present.

Appendix G. Acknowledgement

This research was funded by National Science and Technology Council of Taiwan (NSTC113-2221-E-002-175-MY3).

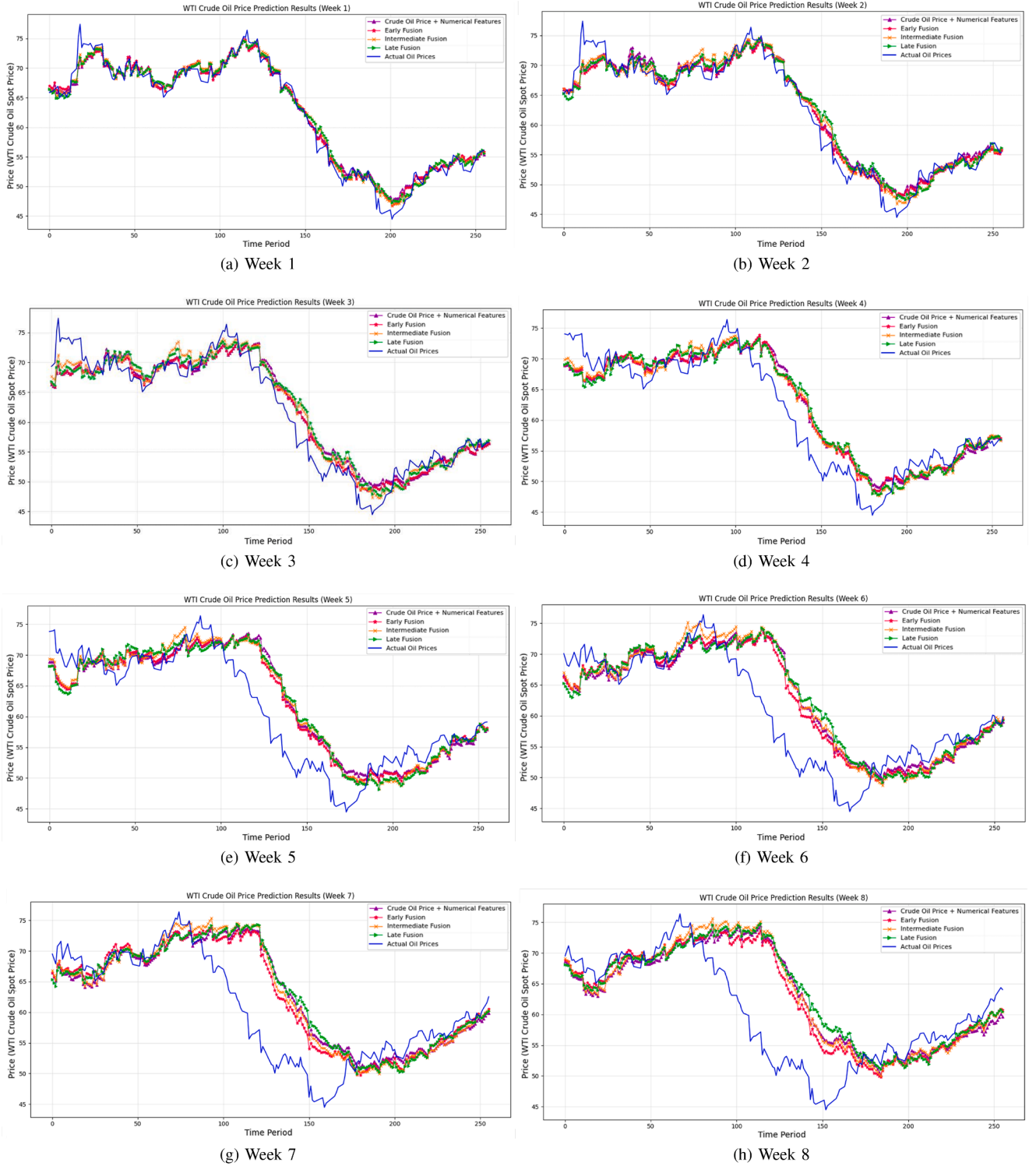


Fig. 7. Best baseline vs. fusion strategies across Weeks 1–8.

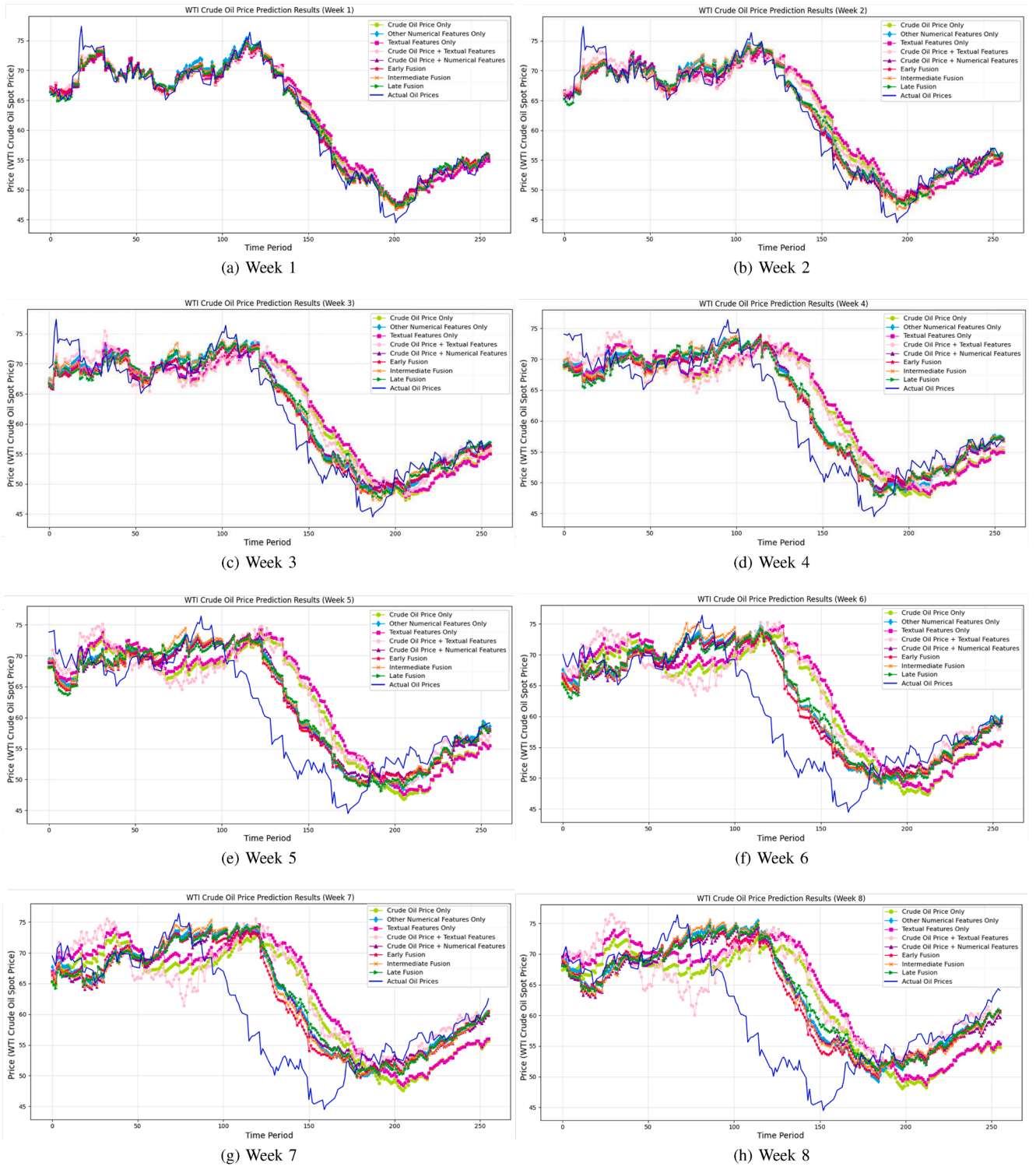


Fig. 8. All baselines vs. fusion strategies across Weeks 1–8.

References

- Abdollahi, H. (2020). A novel hybrid model for forecasting crude oil price based on time series decomposition. *Applied Energy*, 267, 115035. <https://www.sciencedirect.com/science/article/pii/S030626192030547X>. <https://doi.org/10.1016/j.apenergy.2020.115035>
- Bai, Y., Li, X., Yu, H., & Jia, S., et al. (2022). Crude oil price forecasting incorporating news text. *International Journal of Forecasting*, 38(1), 367–383. <https://www.sciencedirect.com/science/article/pii/S0169207021001060>. <https://doi.org/10.1016/j.ijforecast.2021.06.006>
- Bilal, M., Aamir, M., Abdullah, S., & Khan, F., et al. (2024). Impacts of crude oil market on global economy: Evidence from the Ukraine-Russia conflict via fuzzy models. *Heliyon*, 10(1). [https://www.cell.com/heliyon/abstract/S2405-8440\(23\)11082-6](https://www.cell.com/heliyon/abstract/S2405-8440(23)11082-6). <https://doi.org/10.1016/j.heliyon.2023.e23874>
- Chen, R., Deng, T., Huang, S., & Qin, R., et al. (2015). Optimal crude oil procurement under fluctuating price in an oil refinery. *European Journal of Operational Research*, 245(2), 438–445. <https://www.sciencedirect.com/science/article/pii/S0377221715001952>. <https://doi.org/10.1016/j.ejor.2015.03.002>
- Ding, L., Zhao, H., & Zhang, R., et al. (2024). Predicting multi-frequency crude oil price dynamics: Based on MIDAS and STL methods. *Energy*, 313, 134003. <https://www.sciencedirect.com/science/article/pii/S0360544224037812>. <https://doi.org/10.1016/j.energy.2024.134003>
- Fan, Y., Liang, Q., & Wei, Y.-M., et al. (2008). A generalized pattern matching approach for multi-step prediction of crude oil price. *Energy Economics*, 30(3), 889–904. <https://www.sciencedirect.com/science/article/pii/S0140988306001289>. <https://doi.org/10.1016/j.eneco.2006.10.012>
- Fang, T., Zheng, C., & Wang, D., et al. (2023a). Forecasting the crude oil prices with an EMD-ISBM-FNN model. *Energy*, 263, 125407. <https://www.sciencedirect.com/science/article/pii/S0360544222022897>. <https://doi.org/10.1016/j.energy.2022.125407>
- Fang, Y., Wang, W., Wu, P., & Zhao, Y., et al. (2023b). A sentiment-enhanced hybrid model for crude oil price forecasting. *Expert Systems with Applications*, 215, 119329. <https://www.sciencedirect.com/science/article/pii/S0957417422023478>. <https://doi.org/10.1016/j.eswa.2022.119329>
- Han, P., Chen, H., Rasool, A., Jiang, Q., & Yang, M., et al. (2025). MFB: A generalized multimodal fusion approach for bitcoin price prediction using time-lagged sentiment and indicator features. *Expert Systems with Applications*, 261, 125515. <https://www.sciencedirect.com/science/article/pii/S0957417424023820>. <https://doi.org/10.1016/j.eswa.2024.125515>
- Hangloo, S., & Arora, B. (2025). Multimodal fusion techniques: Review, data representation, information fusion, and application areas. *Neurocomputing*, 649, 130827. <https://www.sciencedirect.com/science/article/pii/S09525231225014997>. <https://doi.org/10.1016/j.neucom.2025.130827>
- Hsu, Y.-C., Hung, Y.-H., & Lee, C.-Y., et al. (2025). Robust ensemble forecasting and deep reinforcement learning for energy management on islanded microgrids. *International Journal of Electrical Power & Energy Systems*, 173, 111405. <https://www.sciencedirect.com/science/article/pii/S0142061525009536>. <https://doi.org/10.1016/j.ijepes.2025.111405>
- Huang, Y., & Deng, Y. (2021). A new crude oil price forecasting model based on variational mode decomposition. *Knowledge-Based Systems*, 213, 106669. <https://www.sciencedirect.com/science/article/pii/S095070512030798X>. <https://doi.org/10.1016/j.knosys.2020.106669>
- Kaplan, H., Mundani, R.-P., Rölke, H., & Weichselbraun, A., et al. (2026). CrudeBERT: Applying economic theory towards fine-tuning transformer-based sentiment analysis models to the crude oil market. In *25th International conference on enterprise information systems* (pp. 324–334). <https://www.scitepress.org/Link.aspx?doi=10.5220/0011749600003467>
- Lee, C.-Y., Chou, B.-J., & Huang, C.-F. (2022). Data science and reinforcement learning for price forecasting and raw material procurement in petrochemical industry. *Advanced Engineering Informatics*, 51, 101443. <https://www.sciencedirect.com/science/article/pii/S1474034621001956>. <https://doi.org/10.1016/j.aei.2021.101443>
- Lee, C.-Y., & Yang, S.-H. (2023). Graph spatio-temporal networks for manufacturing sales forecast and prevention policies in pandemic era. *Computers & Industrial Engineering*, 182, 109413. <https://www.sciencedirect.com/science/article/pii/S0360835223004370>. <https://doi.org/10.1016/j.cie.2023.109413>
- Li, X., Shang, W., & Wang, S., et al. (2019). Text-based crude oil price forecasting: A deep learning approach. *International Journal of Forecasting*, 35(4), 1548–1560. <https://www.sciencedirect.com/science/article/pii/S0169207018301110>. <https://doi.org/10.1016/j.ijforecast.2018.07.006>
- Li, Y., Jiang, S., Li, X., & Wang, S., et al. (2021). The role of news sentiment in oil futures returns and volatility forecasting: Data-decomposition based deep learning approach. *Energy Economics*, 95, 105140. <https://www.sciencedirect.com/science/article/pii/S0140988321000451>. <https://doi.org/10.1016/j.eneco.2021.105140>
- Liu, Z., Huang, D., Huang, K., Li, Z., & Zhao, J., et al. (2021). FinBERT: A pre-trained financial language representation model for financial text mining. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence IJCAI'20* (pp. 4513–4519). Yokohama, Yokohama, Japan. <https://doi.org/10.5555/3491440.3492062>
- Patidar, A. K., Jain, P., Dhasmana, P., & Choudhury, T., et al. (2024). Impact of global events on crude oil economy: A comprehensive review of the geopolitics of energy and economic polarization. *GeoJournal*, 89(2), 50. <https://doi.org/10.1007/s10708-024-11054-1>
- Selmi, R., Hammoudeh, S., & Wohar, M. E., et al. (2023). What drives most jumps in global crude oil prices? Fundamental shortage conditions, cartel, geopolitics or the behaviour of financial market participants. *The World Economy*, 46(3), 598–618. <https://onlinelibrary.wiley.com/doi/abs/10.1111/twec.13323>. <https://doi.org/10.1111/twec.13323>
- Tang, L., Zhang, C., Li, L., & Wang, S., et al. (2020). A multi-scale method for forecasting oil price with multi-factor search engine data. *Applied Energy*, 257, 114033. <https://www.sciencedirect.com/science/article/pii/S0306261919317209>. <https://doi.org/10.1016/j.apenergy.2019.114033>
- Windsor, E., & Cao, W. (2022). Improving exchange rate forecasting via a new deep multimodal fusion model. *Applied Intelligence*, 52(14), 16701–16717. <https://doi.org/10.1007/s10489-022-03342-5>
- Wu, B., Wang, L., Lv, S.-X., & Zeng, Y.-R., et al. (2021). Effective crude oil price forecasting using new text-based and big-data-driven model. *Measurement*, 168, 108468. <https://www.sciencedirect.com/science/article/pii/S0263224120309982>. <https://doi.org/10.1016/j.measurement.2020.108468>
- Yu, L., Wang, Z., & Tang, L., et al. (2015). A decomposition-ensemble model with data-characteristic-driven reconstruction for crude oil price forecasting. *Applied Energy*, 156, 251–267. <https://www.sciencedirect.com/science/article/pii/S0306261915008545>. <https://doi.org/10.1016/j.apenergy.2015.07.025>
- Yu, Y., Si, X., Hu, C., & Zhang, J., et al. (2019). A Review of recurrent neural networks: LSTM cells and network architectures. *Neural Computation*, 31(7), 1235–1270. https://doi.org/10.1162/neco_a_01199
- Zhang, X., Cheng, S., Zhang, Y., Wang, J., & Wang, S., et al. (2024). An attention-PCA based forecast combination approach to crude oil price. *Expert Systems with Applications*, 240, 122463. <https://www.sciencedirect.com/science/article/pii/S0957417423029652>. <https://doi.org/10.1016/j.eswa.2023.122463>
- Zhu, S., Xu, M., Wu, J., Wang, Y., Jiang, X., Huang, Z., Wang, Y., & Zhu, Y., et al. (2025). A study on crude oil price forecasting model integrating CEEMDAN-VMD multiscale decomposition with CNN-BiLSTM. *Results in Engineering*, 27, 106391. <https://www.sciencedirect.com/science/article/pii/S2590123025024612>. <https://doi.org/10.1016/j.rineng.2025.106391>